

Matthew Frazier

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Senior Software Engineer with 10+ years of experience in data, analytics, and distributed systems, with a focus on AI-driven and LLM-based systems. Proven track record designing scalable microservices, REST API integrations, and high-volume data pipelines supporting large-scale applications. Delivered end-to-end solutions across backend, data, and full-stack layers, with additional expertise in AI-driven systems including LLM-powered analytics and retrieval pipelines. Ph.D. in Computer and Information Sciences with research in conversational agents intersecting HCI and applied AI.

CORE TECHNICAL SKILLS

Languages: Python, SQL, TypeScript/JavaScript, Node.js

AI/ML Systems: LLMs, retrieval-augmented generation (RAG), LangChain, LangGraph, CrewAI, Qdrant, ChromaDB, Langfuse, Snowflake Cortex, OpenAI, MLflow, AWS SageMaker, prompt engineering (CoT, ReAct, few-shot), Vector Databases, Inference Pipelines, human-in-the-loop (HITL), LLM-as-a-judge

Backend & APIs: REST, gRPC, webhooks, GraphQL, microservices, system design, algorithms, data structures

Data Engineering & ETL Pipelines: Airflow (MWAA), Argo Workflows, Adverity, data ingestion, data modeling

Cloud & Infrastructure: AWS (EKS, ECS, EC2, Lambda, S3, SQS, SNS, RDS, IAM, Route 53, SM, SSM), Kubernetes, Docker, Serverless

Databases & Storage: PostgreSQL, Redis, NoSQL databases, Redshift, Snowflake

Event-Driven Systems: AWS Lambda, SQS (FIFO), SNS, asynchronous processing, streaming workflows

Frontend & Visualization: React, Next.js, Streamlit, data visualization

DevOps & CI/CD: CircleCI (Orb), GitHub Actions, containerization, deployment pipelines, Helm, environment management (dev/local, staging, production)

Testing: Pytest, Jest, Vitest, Storybook, Test-Driven Development (TDD)

Monitoring & Observability: Datadog, Rollbar, AWS Cloudwatch

Data & Analytics Tools: Tableau, Redash, Jupyter Notebooks, Sigma

Security & Auth: OAuth, Auth0, Okta

Integrations: Google Ads, Meta, The Trade Desk (TTD), YouGov, Salesforce, SFTP

PROFESSIONAL EXPERIENCE (select)

Senior Software Engineer (Data Science, Scaled Media Solutions) | Goodway Group | New York, N.Y. 2021 - Present

- Designed and implemented AI-driven analytics systems within the **GOES™** (Goodway Outcome Engineering System) platform, developing LLM-based systems for natural language query (NLQ), RAG, and hierarchical insight generation. Built production-grade AI workflows integrated with vector search, stateful memory, prompt engineering (CoT, ReAct, few-shot), and streaming inference pipelines via REST APIs to support real-time decision-making and analytics systems.
- Improved production stability of **RealValue™**, a high-throughput distributed optimization system (bid factoring, bid range, combinatorial and hyperlocal optimization) processing **40M+** impression-level decisions daily across campaign datasets, built on microservices, ETL pipelines, and DSP integrations (e.g., TTD). Re-architected error handling (Rollbar, Datadog) and validation logic within pipeline workflows, increasing system reliability while reducing production errors by **90%**.
- Integrated backend microservices and REST APIs across **5+** external platforms (TTD, Bing, YouGov, Google Ads, Meta). Built high-volume ETL data pipelines (Airflow, AWS MWAA, Argo Workflows, Redshift, Snowflake, S3) supporting large-scale data and analytics systems. Led DevOps initiatives and co-led AI initiatives with Data Science leadership, driving best practices and cross-team MLOps governance alignment across platform and AI systems.
- Architected full-stack and event-driven distributed systems, including AWS Lambda, SNS, SQS (FIFO), and S3 workflows to automate YouGov analysis pipelines, replacing manual Jupyter notebook execution with asynchronous processing. Deployed frontend (React/Next.js, Streamlit) and data visualization interfaces for real-time insights.

Senior Technical Operations Engineer (Business Intelligence) | Goodway Group | New York, N.Y. 2018 - 2021

- Owned data reliability, integrity, and availability of marketing analytics systems. Managing end-to-end data operations and reporting infrastructure (Airflow, Redshift, S3, Tableau) under strict daily data freshness SLAs.
- Owned org-wide incident response, root-cause analysis, postmortem, and data discrepancy triage across ETL pipeline and reporting systems. Adopted preventative fixes and monitoring workflows that improved system stability and reduced recurring issues.

Principal Campaign Operations Engineer | Goodway Group | New York, N.Y. 2014 - 2018

- Launched **COLT** (Cacidi On-Line Tool), a web-based automation system augmenting Adobe InDesign templates that parameterized ad copy and legal logic. Automated campaign production, reducing creative approval time by approximately **80% (from 10-12 business days to 2 business days)** and enabling scalable execution across multi-location automotive clients (e.g., Chevrolet).
- Executed, QA'd, and reported on programmatic campaigns across platforms (TTD, Google Ads, YouTube, Meta), collaborating with ad operations and media trading to improve data-driven campaign pacing and performance.

Senior Search Engine Optimization (SEO) Strategist | IONOS (1&1 Internet) | Chesterbrook, PA 2012 - 2014

- Established and analyzed SEO and Search Engine Marketing (SEM) strategies through copy and creative variation A/B testing, metatag optimization, user engagement behavior analysis (CTR, time-on-page), and keyword analysis, improving organic visibility and traffic across **300+** small business websites in US, UK, and Canadian markets (Google Analytics, SEMrush).

RESEARCH & TEACHING EXPERIENCE

PhD Researcher & Instructor | University of Delaware | Newark, DE 2018 - 2024

- Led research on conversational agents for computer science education, resulting in peer-reviewed publications. Engineered and iterated on LLM-based prototype systems using prompt engineering techniques (e.g., RAG, CoT, few-shot) to promote AI learning systems that guide Socratic reasoning rather than direct answer generation.
- Instructed undergraduate computer science courses. Created curriculum and taught programming, data structures, and computer science ethics.
- **Research Expertise:** User Interviews, Survey Research, Usability Testing, Concept Testing, Experimental Design, Participatory Design.

Graduate Assistant & Researcher | West Chester University of Pennsylvania | West Chester, PA 2016 - 2017

- Evaluated data structures assignments (Java), assessing correctness, efficiency, and algorithmic approach.
- Developed an object detection and identity matching application (computer vision - Python, OpenCV) that analyzes university parking lot videos to detect open parking spots for student lot availability.

EDUCATION

University of Delaware | Newark, DE

Ph.D., Computer and Information Sciences (Magna Cum Laude)

2024

West Chester University of Pennsylvania | West Chester, PA

M.Sc., Computer Science (Cum Laude)

2017

B.Sc., Marketing

2010

PUBLICATIONS (select)

- Customizing ChatGPT to Help Computer Science Principles Students Learn Through Conversation (ITiCSE 2024)
- Investigating User Perceptions of Conversational Agents for Software-related Exploratory Web Search (ICSE 2022)